Home Work No. 01

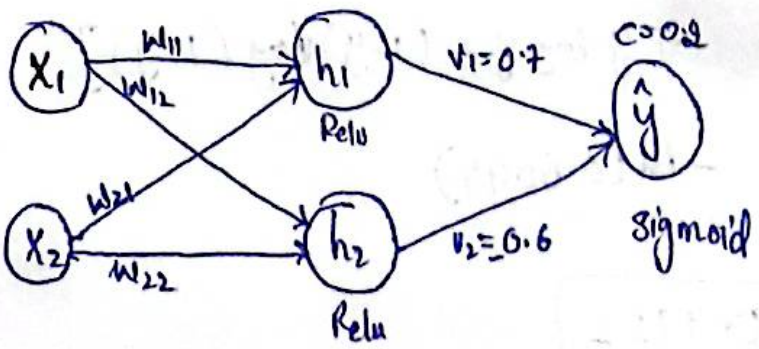
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Section :- V3
Course :- DL & NN
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Question No.1 :-

Given data :- $X_1 = 1.0$, $X_2 = 0.5$, $y = 1$, $\eta = 0.5$

$W_{11} = 0.4$	$W_{21} = -0.3$	$V_1 = 0.7$
$W_{12} = -0.5$	$W_{22} = 0.2$	$V_2 = -0.6$
$b_1 = 0.1$	$b_2 = 0.1$	$c = 0.2$

Solution :-



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Part (a) Forward Pass :-

Hidden Neuron 1 :-

$$z_1 = w_{11} x_1 + w_{21} x_2 + b_1$$

$$z_1 = 0.4(1) + (-0.3)(0.5) + 0.1 = 0.35$$

$$h_1 = \text{Relu}(0.35) = \boxed{0.35}$$

Hidden Neuron 2 :-

$$z_2 = w_{12} x_1 + w_{22} x_2 + b_1$$

$$z_2 = -0.5(1) + 0.2(0.5) + 0.1 = -0.30$$

$$h_2 = \text{Relu}(-0.30) = \boxed{0}$$

Output Neuron :-

$$z_o = v_1 h_1 + v_2 h_2 + c$$

$$z_o = 0.7(0.35) - 0.6(0) + 0.2 = 0.445$$

$$\hat{y} = \sigma(0.445) = \frac{1}{1+e^{-(0.445)}} = 0.6094$$

$$\boxed{\hat{y} = 0.6094}$$

Part b :- loss = ?

$$BCE = L = -[y \log \hat{y} + (1-y) \log(1-\hat{y})]$$

$$L = -\ln(0.6094)$$

$$\boxed{L = 0.4952}$$

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Part C:- Gradient of All Nine Parameters :-

For Sigmoid output with BCE :-

$$\frac{\partial L}{\partial z_o} = \hat{y} - y$$

$$\delta_o = 0.6094 - 1 = \boxed{-0.3906}$$

Output layer gradient :

$$\frac{\partial L}{\partial v_1} = \delta_o \cdot h_1 = (-0.3906)(0.35) = \boxed{-0.1367}$$

$$\frac{\partial L}{\partial v_2} = \delta_o \cdot h_2 = (-0.3906)(0) = \boxed{0}$$

$$\frac{\partial L}{\partial c} = \delta_o \cdot 1 = (-0.3906) = \boxed{-0.3906}$$

For hidden layer Neuron 1 :-

$$\delta_1 = \delta_o v_1 = -0.3906\,(0.7) = \boxed{-0.2734}$$

$$\frac{\partial L}{\partial w_{11}} = \delta_1 x_1 = (-0.2734)(1) = \boxed{-0.2734}$$

$$\frac{\partial L}{\partial w_{21}} = \delta_1 x_2 = (-0.2734)(0.5) = \boxed{-0.1367}$$

$$\frac{\partial L}{\partial b_1} = \delta_1 = (-0.2734) = \boxed{-0.2734}$$

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For hidden neuron 2 → $z_2 < 0$ → so Relu is zero

$$\delta_2 = \delta_0 V_2(0) = 0$$

So: $\frac{\partial L}{\partial W_{12}} = 0$ $\frac{\partial L}{\partial W_{22}} = 0$ $\frac{\partial L}{\partial b_2} = 0$

Part = d :- ONE gradient descent update :-

Sol $\theta_{new} = \theta_{old} - \eta \cdot \frac{\partial L}{\partial \theta}$

✱ Now put one parameter $\frac{\partial L}{\partial V_1} = -0.1367$

$$V_{new} = 0.7 - 0.5(-0.1367)$$

$$\boxed{V_{new} = 0.76}$$

✱ $L_{new} = -\ln(0.76) \approx 0.341$

✱ So Now loss is decrease from 0.4952 to 0.3410

Part e :- Leaky Relu

Given → $\alpha = 0.01$

$$\delta_2 = \delta_0 V_2(0.01) = (-0.3906)(-0.6)(0.01) = \boxed{0.00234}$$

$$\frac{\partial L}{\partial W_{21}} = \delta_2 \cdot x_1 = (-0.00234)(1) = \boxed{0.00234}$$

$$\frac{\partial L}{\partial W_{22}} = \delta_2 \cdot x_2 = (-0.00234)(0.5) = \boxed{0.001172}$$

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Question.02 :- Activation function, gradient flow, & Softmax.

Sol (Part=a)

Sigmoid $= \frac{1}{1+e^{-z}} \rightarrow (0, 1)$

Tan h $= \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}} \rightarrow (-1, 1)$

Relu $= \max(0, z) \rightarrow (0, \infty)$

Leaky Relu $= \max(0.01z, z) \rightarrow (-\infty, \infty)$

ELU $= z$ if $z > 0$ else $e^{-z} - 1 \rightarrow (-1, \infty)$

Swish $= z \cdot \sigma(z) \longrightarrow \approx [-0.28, \infty)$

$\sigma'(z) = \sigma(z)(1 - \sigma(z))$

$\tanh'(z) = 1 - \tanh^2(z)$

$\text{ReLU}'(z) = \begin{cases} 1 & z > 0 \\ 0 & z \leq 0 \end{cases}$

$\text{Leaky ReLU}'(z) = \begin{cases} 1 & z > 0 \\ 0.01 & z \leq 0 \end{cases}$

	$f(-2)$	$f'(-2)$	$f(0)$	$f'(0)$	$f(2)$	$f'(2)$
sigmoid $\sigma(z)$	0.1192	0.1050	0.5000	0.2500	0.8808	0.1050
Tan h	-0.9640	0.0707	0	1.0000	0.9640	0.0707
Relu	0	0	0	0	2	1
Leak Relu	-0.0200	0.0100	0	0.0100	2	1
ELU	-0.8647	0.1353	0	1.000	2	1
Swish	-0.2384	-0.093	0	0.5000	1.7616	1.0908

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Part b :- Vanishing & exploding gradients :-

$$\frac{\partial a_6}{\partial a_0} = \prod_{k=1}^{6} w_k f'(z_k)$$

(1) Sigmoid, $z_k = 0$, $w = 0.8$:

$$((0.8) \times (0.25))^6 = \boxed{0.000064}$$

(2) Tanh $z_k = 0$ $w = 0.8$

$$(0.8 \times 1)^6 = \boxed{0.262144}$$

(3) ReLU $z_k > 0$, $w = 0.8$

$$(0.8 \times 1)^6 = \boxed{0.262144}$$

(4) ReLU $z_k > 0$ $w = 1.5$

$$(1.5 \times 1)^6 = \boxed{11.390625}$$

Part C :- Softmax & Cross-entropy :-

$$z = (2, 1, -0.5), \quad y = (0, 1, 0)$$

softmax probabilities are:

$$P_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

$$P \approx (0.6897, 0.2537, 0.0566)$$

Since class 2 the true class

$$L = -\ln(P_2) = -\ln(0.2537)$$

$$\boxed{L = 1.3715}$$

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For softmax with cross-entropy.

$$\frac{\partial L}{\partial z} = p - \hat{y}$$

$p = (0.6897, 0.2537, 0.0566)$

$y = (0, 1, 0)$

$$\boxed{\frac{\partial L}{\partial z} = (0.6897, -0.7463, 0.0566)}$$

$0.6897 - 0 = 0.6897$

$(0.6897 - 0)(0.5) = 0.3448$

$(-0.7463)(0.5) = -0.373$

$(0.0566 - 0)(0.5) = 0.0283$

Give $h = (0.5, 1, 0)^T$ & $z = Wh + b$

$$\frac{\partial L}{\partial W} = (p - y)h^T$$

$$\underset{3\times 2}{\frac{\partial L}{\partial W}} \approx \begin{bmatrix} 0.3448 & 0.6897 \\ 0.373 & -0.7463 \\ 0.0283 & 0.0566 \end{bmatrix}$$

Question no. 3 :- Gradient-based optimizer :

Given:- $L(w_1, w_2) = \frac{1}{2} w_1^2 + 10 w_2^2$, $w^{(0)} = (2, 1)$

Part a :- Gradient, Hessian & condition number.

$$\nabla L = \begin{bmatrix} w_1 \\ 20 w_2 \end{bmatrix}$$

$2 \times \frac{1}{2} w_1 + 20 w_2$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 20 \end{bmatrix}$$

Eigen value is 1 to 20 $\Rightarrow K = \frac{20}{1} = \boxed{20}$

the condition number is

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For gradient descent to converge in both directions

$$0 < \eta < 2/\lambda_{max} = \frac{2}{20}$$

$$\boxed{0 < \eta < 0.1}$$

Part b :- Two Gradient descent step

$$w^{(t+1)} = w^t - \eta \nabla L(w^{(t)})$$

First Step :- $W^{(1)} = (2,1) - 0.15(2,20) = \boxed{(1.7, 2)}$

$$L(w^{(1)}) = \frac{1}{2}(1.7)^2 + 10(-2)^2 = \boxed{41.445}$$

Second step :- $\nabla L(w^{(1)}) = (1.7, -40)$

$$w^{(2)} = (1.7, -2) - 0.15(1.7, -40) = \boxed{(1.445, 4)}$$

$$L(w^{(2)}) = \frac{1}{2}(1.445)^2 + 10(4)^2 \approx \boxed{160.044}$$

Part C Two Steps $\eta = 0.25$

$$w^{(1)} = (2,1) - 0.25(2,80) = (1.5, -4)$$

$$L = \frac{1}{2}(1.5)^2 + 10(16) = 160 + 1.125 = \boxed{161.125}$$

$$w^{(2)} = (1.5, -4) - 0.25(1.5, -80) = (1.125, 16)$$

$$L = \frac{1}{2}(1.125)^2 + 10(256) = \boxed{2560.633}$$

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Part=d

Momentum, $\beta = 0.9$, $\eta = 0.05$

$$V_t = \beta V_{t-1} + \nabla L(w^{t-1})$$

$$w^t = w^{t-1} - \eta V_t$$

Sol Step 1:-

$V_1 = (2, 20)$ $\quad w^{(1)} = (2, 1) - 0.05(2, 20)$

$$\boxed{w^{(1)} = (1.9, 0)}$$

Step 2:- $\quad \nabla L(w^{(1)}) = (1.9, 0)$

$$V_2 = 0.9(2, 20) + (1.9, 0)$$

$$\boxed{V_2 = 3.7, 18}$$

$$w^{(2)} = (1.9, 0) - 0.05(3.7, 18)$$

$$\boxed{w^{(2)} = (1.715, -0.9)}$$

Part=e Adam

Given:- $\eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.99$, $\epsilon = 10^{-8}$

Sol $\quad m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$

$$S_t = \beta_2 S_{t-1} + (1 - \beta_2) g_t^2$$

Bias correction:-

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \qquad \hat{S}_t = \frac{S_t}{1 - \beta_2^t}$$

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(10)

$$w_t = w_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{s}_t} + \epsilon}$$

Step	g_t	m_t	s_t	$\hat{m}_t$	$\hat{s}_t$	w_t
1	(2,20)	(0.2,2)	(0.004,0.4)	(2,20)	(4,400)	(1.9, 0.9)
2	(1.9,18)	(0.37,3.6)	(0.0076,0.723)	(1.9,18.9)	(3.8,361.9)	(1.8, 0.802)

Part (f) Mini_batch updates :-

updates per epoch = $\frac{60000}{128} \approx$ **469**

For 15 epoch :-

$469 \times 15 =$ **7035 updates**

Advantage of Mini-batch & full batch GD :-

(★) Mini batch SGD is that it uses less memory per updates

(★) Full batch GD can update parameter more frequently.

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Question 04 :— Initialization & Regularization.

Part a A fully connected layer maps $n_{in} = 512$ inputs to $n_{out} = 256$ outputs

(i) $\sqrt{\frac{6}{n_{in}+n_{out}}}$ $\sqrt{\frac{2}{n_{in}+n_{out}}}$ $\sqrt{\frac{2}{n_{in}}}$

$= \sqrt{\frac{6}{512+256}}$ $\sqrt{\frac{2}{512+256}}$ $\sqrt{\frac{2}{512}}$

$= \boxed{0.0884}$ $\boxed{0.0510}$ $\boxed{0.0625}$

(II) Variance accross layer

$\sigma_w = 0.01$

$Var(Z) = n_{in}\sigma_w^2$

$= 512 \cdot (0.01)^2$

$\boxed{Var(Z) = 0.0512}$ → one layer

After 10 layer → $(0.0512)^{10} \simeq \boxed{1.22 \times 10^{-13}}$

(III) If all weights begin at zero, neuron in the same layer have identical weights and recieves same potential/ identical gradients. This continuous learning the features. This is called Symmetry problem

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Part b Weight decay

Given $w = 0.8$, data-loss gradient 0.1, $\lambda = 0.01$

⇒ Without Regularization :-

$$w_{new} = 0.8 - 0.1(0.1) = \boxed{0.7900}$$

⇒ With L_2 penalty λw^2, the additional gradient $2\lambda w = 0.016$

$$w_{new} = 0.8 - 0.1(0.1) + 0.016 = \boxed{0.7884}$$

⇒ With L_1 penalty $\lambda |w|$ the additional gradient 0.01

$$w_{new} = 0.8 - 0.1(0.11) = \boxed{0.7890}$$

L_1 regularization can make weight exactly zero through thresholding; L_2 usually shrinks weights without setting them exactly to zero

Part c Inverted dropout :-

Given $a = (2, 0, 1.5, 3, 1)$ $p = 0.8$

$m = (1, 1, 0, 1, 1)$

Training output = $a \cdot \frac{m}{p} = \boxed{(2.5, 0, 0, 3.75, 1.25)}$

Mean of traing output = $\frac{2.5+0+0+3.75+1.25}{5} = \boxed{1.5}$

original mean = $\frac{2+0+1.5+3+1}{5} = \boxed{1.5}$

(*) At test time, dropout is turn off. Inverted dropout already scales retained during activation.

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Part d

Batch Normalization :-

Batch : (2, 4, 6, 8)

Mean :

$$\mu = \frac{2+4+6+8}{4}$$

$$\boxed{\mu = 5}$$

Biased variance :-

$$\sigma^2 = \frac{(2-5)^2+(4-5)^2+(6-5)^2+(8-5)^2}{4}$$

$$\boxed{\sigma^2 = 5}$$

Normalized values :

$$\hat{x}_i = \frac{x_i - 5}{\sqrt{5+10^{-5}}}$$

$$x_1 = \frac{2-5}{\sqrt{5+10^{-5}}} = -1.3416$$

$$x_2 = \frac{4-5}{\sqrt{5+10^{-5}}} = -0.4472$$

$$x_3$$

$$\boxed{\hat{x}_i \approx (-1.3416, -0.4472, 0.4472, 1.3416)}$$

With $\gamma = 1.5$, $\beta = 0.5$

$$y_i = \gamma \hat{x}_i + \beta$$

$$y_i = 1.5(-1.3416) + 0.5 = -1.5125$$

$$y_2 = 1.5(-0.4472) + 0.5 = -0.1708$$

$$\boxed{y \approx (-1.5125, -0.1708, 1.1708, 2.5125)}$$

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